

GABRIELA BIEDA

Bioinformatics | Computational Chemist | Data Analyst

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PROFESSIONAL SUMMARY

MSc graduate in Bioinformatics with over two years of academic and project experience in computational biology, molecular modelling and data analysis. Strong skills in Python, molecular dynamics, machine learning and quantitative analysis of biomolecular systems. Completed research projects involving DNA deamination modelling, protein–ligand simulations and reproducible bioinformatics workflows.

AREAS OF EXPERTISE

Professional skills: Bioinformatics | computational biology | molecular modelling | molecular dynamics | molecular docking | protein-ligand interaction analysis | computational chemistry | machine learning | statistical data analysis | biological sequence analysis | data visualization

Technical skills: Python | R | NumPy | pandas | scikit-learn | TensorFlow | SQL | GROMACS | NAMD | VMD | PyMOL | Gaussian | Linux | Git | HPC | PLGrid | AutoDock | Maestro | Chimera | Avogadro | Molden

PROFESSIONAL EXPERIENCE

Organic Chemist, Selvita Chemistry Academy Selvita | Cracow, Poland Nov. 2024 - Jan. 2025

- Performance of basic organic synthesis reactions in the pharmaceutical industry, including amide formation, Wittig, Buchwald and Pictet-Spengler reactions
- Operation of laboratory equipment and analysis of ^1H NMR and LC-MS results
- Literature research related to the performed chemical reactions
- Selection and optimization of reaction conditions, including temperature, reaction time and reagents

Cosmetic Formulation and Quality Control Intern Euphora Paese | Jawornik, Poland Jul. 2024 - Aug. 2024

- Production of cosmetic formulations at the laboratory scale
- Analysis of raw materials and manufactured batches, including final product quality verification
- Product labeling in accordance with production requirements
- Packaging of finished goods

RESEARCH EXPERIENCE

MSc Thesis Mar. 2025 - Jul. 2026

“Spontaneous deamination of cytosine in DNA. Modelling the impact of nucleotide sequence”

- Preparation of computational DNA models with different local nucleotide sequences
- Quantitative analysis of molecular interactions and transition-state stabilization
- Processing of structured computational datasets and comparison of molecular variants
- Identification of sequence-dependent patterns in complex numerical results
- Development of reproducible workflows for data analysis and visualization
- Automation of selected stages of data processing and result preparation

Student Researcher, FutureLab Research Projects, Bioactive Lab Scientific Circle Dec. 2022 - Mar. 2025

“TryPto Holder - A new method for treating pancreatic cancer using compounds that inhibit serotonin biosynthesis by blocking the TPH1 enzyme” and “New ligands of the 5-HT6 receptor from the group of sulfonamide derivatives of cyclic aryloguanidines in the treatment of glioblastomas” and Sulfonamide derivatives of 1,4-dihydroquinazolin-2-amine as a new scaffold for selective 5-HT6R ligand design”

- Preparation and standardization of protein and ligand structures for in silico analysis
- Performance of molecular docking and Induced Fit Docking to model protein-ligand binding
- Preparation of protein–ligand complexes for molecular dynamics simulations
- Analysis of binding poses, scoring metrics and key molecular interactions
- Evaluation of complex stability based on RMSD trends, contact persistence and molecular dynamics outputs
- Identification of structural interaction patterns to support candidate ranking and selection
- Use of HPC resources, including the PLGrid infrastructure at ACK Cyfronet AGH, and organization of computational results
- Processing and visualization of simulation data in Python to identify patterns and support data-driven analysis

Member, Event Coordinator and Mentor Independent Students' Association, NZS

Feb. 2022 - Jul. 2026

- Coordination of student initiatives, including “Podaj Łapę” and “Wampiriada”, covering event planning, promotion
- Collaboration with team members in event organization and task allocation
- Mentoring of new members and supporting their integration into the organization
- Development of teamwork, communication, leadership and event coordination skills

Education

Wrocław University of Science and Technology

Mar. 2025 - Jul. 2026

MSc studies in Bioscience,Bioinformatics (in English)

Cracow University of Technology

Oct. 2021 - Feb. 2025

BSc studies in Chemical Technology, Light Organic Chemistry

Languages

Polish

Native

English

C1

Spanish

B1

German

A1

Key Achievements

- Eastern European Bioinformatics and Computational Genomics Workshop (EEBG) | 2026
- Unite! Research School, Politecnico di Torino | 2026
- Rector’s Scholarship for outstanding academic achievements, awarded in the academic years 2022-2026
- Completed “The Complete Python Bootcamp: From Zero to Hero” | 2025
- Santander Scholarship for students of the Krakow University of Technology | 2024
- Second place in the Departmental Scientific Circle Session | 2024
- Attended the “Frontiers in Medicinal Chemistry” Conference, Munich, 2024
- Completed “Bioinformatics: Learn Docking and Molecular Dynamics Simulation” | 2024
- Merck Laboratory Training Series | 2024
 - Practical Applications of Ultrafiltration in Sample Preparation
 - LC–MS/MS in Modern Analysis of Biological and Food Samples
 - Selection and Proper Use of Analytical- and Chromatography-Grade Reagents
 - Cost-Efficient Operation of Laboratory Equipment, Including Incubators, Drying Ovens, Freezers and Climate Chambers